TUTORIAL
ADVANCED NMRD DATA FITTING WITH FITTEIA AND ONEFIT-ENGINE

#1 Confirm a model fit that was published (fitteia.org)

#2 Fit a multiexponential magnetization decay (fitteia.org)

#3 Fit a series of multiexponential magnetization decays with and estimate the best relative amplitude (OneFit-Engine)

Single molecular rotations/reorientations

The contributions of molecular rotations/reorientations can be treated by the model developed by Nordio *(25)*. Here,

$$(T_1^{-1})_R(\omega_L) = \frac{3}{2}K_{dd}^2 I(I+1)\left[\mathcal{J}_{\mathcal{R}}^{(1)}(\omega_L) + \mathcal{J}_{\mathcal{R}}^{(2)}(2\omega_L)\right] \tag{S2}$$

where the rotational correlation functions used to get spectral densities $\mathcal{J}(\omega)$ are here expressed for cylindrical molecules reorienting in uniaxial mesophases. Following Abragam's notation *(26)* ($m_L \neq 0$):

$$\mathcal{J}_{\mathcal{R}}^{(m_L)}(m_L\omega_L) = \frac{4}{3}(m_L)^2 \sum_{m_M=0}^{2} A^{(m_M)} c(m_L, m_M) \frac{\left(\tau_{m_L m_M}^2\right)^1}{m_L^2\omega_L^2 + \left(\tau_{m_L m_M}^2\right)^{-2}} \tag{S3}$$

with the correlation times τ_{m_L,m_M}^2 given by

$$\left(\tau_{m_L,m_M}^2\right)^{-1} = \frac{\tau_S^{-1}}{\beta_{m_L m_H}^2} + (\tau_L^{-1} - \tau_S^{-1})m_{m_H}^2 \tag{S4}$$

while $c(m_L, m_M)$ and $\beta_{m_L m_M}^2$ can be estimated numerically as functions of the nematic order parameter S. The average molecular geometrical factors $A^{(0)} = \overline{\sum_{j\neq i}\left|d_{0,0}^2(\alpha_{kl})\right|^2/r_{kl}^6}$ and $A^{(m_M)} = 2\overline{\sum_{j\neq i}\left|d_{m_M,0}^2(\alpha_{kl})\right|^2}/r_{kl}^6$ for $m_M \neq 0$ were estimated for 5CB-αd$_2$, assuming the most stretched molecular conformation.

In the isotropic phase the nematic order parameter S is zero and in Eq. (3), $c(m_L, m_M) = 1/5$ and $\beta_{m_L m_H}^2 = 1/6$. τ_L^{-1} and τ_S^{-1} are the rotational correlation times along the long and short molecular axis, respectively.

Correlated rotations/reorientations and short-range order fluctuations

$$J^{(k)}(\omega_0) = \frac{4}{3}c_k \sum_{m=-2}^{2} \frac{\overline{\left|d_{m0}^2(\alpha_{ij})\right|^2}}{r_{ij}^6}\left\langle\left|D_{km}^2(\theta)\right|^2\right\rangle \frac{\tau_{km}^2}{1+\omega_0^2\left(\tau_{km}^2\right)^2} \tag{11.18}$$

$d_{km}^2(\alpha_{ij})$ is the second-rank reduced Wigner rotation matrix and $\langle D_{km}^2(\theta)\rangle = \langle D_{k|m|}^2(\theta)\rangle$ is the second-rank rotation matrix.

$$\frac{\overline{\left|d_{m0}^2(\alpha_{ij})\right|^2}}{r_{ij}^6} = \begin{cases} \overline{\left(3\cos^2\alpha_{ij} - 1\right)^2/\left(4r_{ij}^6\right)}, & m = 0 \\ \overline{3\sin^2 2\alpha_{ij}/\left(8r_{ij}^6\right)}, & m = \pm 1 \\ \overline{3\sin^4\alpha_{ij}/\left(8r_{ij}^6\right)}, & m = \pm 2 \end{cases} \tag{11.19}$$

$$\left\langle\left|D_{k|m|}^2(\theta)\right|^2\right\rangle = \frac{1}{5} + \frac{1}{35}\begin{bmatrix} 10S + 18\langle P_4\rangle - 35S^2 & 5S - 12\langle P_4\rangle & -10S + 3\langle P_4\rangle \\ 5S - 12\langle P_4\rangle & \frac{5}{2}S + 8\langle P_4\rangle & -5S - 2\langle P_4\rangle \\ -10S + 3\langle P_4\rangle & -5S - 2\langle P_4\rangle & 10S + \frac{1}{2}\langle P_4\rangle \end{bmatrix} \tag{11.20}$$